



Review article

A retrospective analysis of 88 anticoagulant rodenticide poisoning cases: Characteristics and forensic implications

Zhonghao Yu^{a,1}, Jiaxin Zhang^{a,1}, Yuhao Yuan^a, Yuan Wang^b, Xiangting Gao^a, Laijing Sun^{c,*}, Yiwu Zhou^{a,*}

^a Department of Forensic Medicine, Tongji Medical College, Huazhong University of Science and Technology, Wuhan 430030, China

^b General Hospital of General Road Street, Dongxihu District, Wuhan 430040, China

^c Shandong Police College, Jinan 250200, China

ARTICLE INFO

Keywords:

Anticoagulant rodenticides

Warfarin

Brodifacoum

Bromadiolone

Poisoning

Case review

ABSTRACT

This study aimed to synthesize and analyze toxicological characteristics from reported cases of human anticoagulant rodenticide poisoning, providing reference data for clinical diagnosis and forensic practices. We systematically identified English-language case reports (published since 2000) of human anticoagulant rodenticide poisoning via PubMed and Web of Science. A retrospective analysis was performed on various characteristics, including year, age, gender, manner of poisoning, latency period, clinical symptoms and examination results, primary treatment methods, pathological changes in fatal cases, types and concentrations of rodenticides, and toxicological analysis methods. A total of 88 poisoning cases were identified, comprising 52 cases of accidental poisoning, 31 cases of suicidal poisoning, 3 cases of unknown poisoning and 2 cases of homicide. The median latency period was 4 days (range: 1 ~ 30). Multi-organ hemorrhage was the predominant clinical manifestation, with hematuria being the most frequently reported symptom ($n = 39$). Coagulation function tests revealed average values of aPTT, PT, and INR of aPTT: 110 (3.71 ~ 212) s, PT: 100 (11.6 ~ 300) s and INR: 9 (0.98 ~ 38.2), respectively, all significantly exceeding normal ranges. Vitamin K₁ administration (intravenous or oral) was the primary treatment. Six fatalities occurred and autopsy findings in three cases primarily indicated multi-organ hemorrhage and necrosis. About the types of rodenticides, a total of 38 cases reported 7 distinct types, with brodifacoum and bromadiolone being the most common, presenting median concentrations of 0.15 µg/mL (0.05 ~ 10) and 0.1215 µg/mL (0.00032 ~ 1.665), respectively, which may be significant for clinical diagnosis and forensic identification. In Conclusion, this retrospective analysis characterized key features of 88 cases of anticoagulant rodenticide poisoning and may provide valuable reference to support clinical management and forensic investigation in suspected poisoning events.

1. Introduction

Anticoagulant rodenticides inhibit the activity of vitamin K epoxide reductase, preventing the reduction of vitamin K from its epoxide form to the active hydroquinone form. This inhibition impacts the synthesis of vitamin K-dependent clotting factors II, VII, IX, and X, as well as proteins C and S, ultimately resulting in coagulation dysfunction [1,2]. While widely employed globally for rodent control, these compounds are also implicated in human poisoning incidents, including fatal cases, as documented in the literature [3,4]. Most existing reports are isolated

case studies, while valuable for describing individual toxicological profiles, inherently lack the scope to provide a comprehensive characterization of the toxicological features of anticoagulant rodenticide poisoning in humans.

Therefore, to address this gap and provide a more accurate depiction of key toxicological characteristics, such as the latency period, clinical manifestations, and toxic concentrations, in human poisoning, we conducted a retrospective analysis of aggregated case reports. We anticipated that the findings from this analysis will provide a valuable foundation to support the clinical diagnosis, treatment, and forensic

* Corresponding authors.

E-mail addresses: zhonghaoyu795@163.com (Z. Yu), 296034673@qq.com (J. Zhang), 362736952@qq.com (Y. Yuan), 286152522@qq.com (Y. Wang), 1404467629@qq.com (X. Gao), sunlaijing@163.com (L. Sun), zhouyiwu@hust.edu.cn (Y. Zhou).

¹ These authors contributed equally to this work.

<https://doi.org/10.1016/j.forensiint.2025.112660>

identification of anticoagulant rodenticide poisoning.

2. Methods

The cases retrieval, screening, and analysis workflow for anticoagulant rodenticide poisoning are detailed in Fig. 1. Descriptive statistics represented by median (range) were performed using GraphPad Prism software (version 8.0.1). Comprehensive characteristics of the 88 cases derived from 74 articles [4–77] are provided in Supplementary Table S1.

3. Results

3.1. Background information

The temporal distribution of the 74 publications revealed progressive growth: 4 (5.4 %) from 2000 ~ 2004, 20 (27.0 %) from 2005 ~ 2009, 23 (31.1 %) from 2010 ~ 2014, 24 (32.4 %) from 2015 ~ 2019, and 17 (23.0 %) from 2020 ~ 2024. Among the 88 cases of poisoning, patients comprised 49 males (55.7 %) and 39 females (44.3 %). Age distribution stratified as follows: Pediatric (0 ~ 17 years): 8 cases (9.1 %), young adults (18 ~ 44 years): 40 cases (45.5 %), middle-aged adults (45 ~ 59 years): 28 cases (31.8 %), geriatric (≥60 years): 12 cases (13.6 %).

3.2. Manner of poisoning, mental condition and synthetic cannabis abuse

Among the 88 poisoning cases, 52 cases were accidental poisoning, 31 cases were suicidal poisoning, 2 cases were homicidal poisoning, and the other 3 cases was unknown. Psychological assessment revealed pre-existing mental health disorders in 29 individuals, among which 28 cases were suicidal poisoning and 1 case was accidental poisoning. Additionally, we found that 11 cases involved the misuse of synthetic cannabis: 9 cases combined accidental poisoning with synthetic cannabis use, and 2 cases involved intentional ingestion alongside synthetic cannabis abuse.

3.3. Latency period

Among the 88 cases, latency period was documented for 53 individuals, which ranged from 1 to 30 days, with a median of 4 days.

3.4. aPTT, PT and INR

Among the 88 poisoned patients, coagulation parameters at initial hospitalization were available for 64 cases in terms of aPTT, 61 cases for PT, and 69 cases for INR. Marked coagulopathy was observed, with mean values substantially exceeding normal limits, which were visually summarized in Fig. 2, confirming the expected anticoagulant effect.

3.5. Clinical symptoms or diagnoses

Of the 88 cases, 7 cases remained asymptomatic throughout the entire process. In the remaining 81 cases, the predominant manifestation was multi-organ hemorrhage, with the most frequent presentations being hematuria (n = 39, 44.3 %), truncal or extremity ecchymoses (n = 35, 39.8 %), gastrointestinal hemorrhage (n = 28, 31.8 %), epistaxis (n = 21, 23.9 %), oral mucosal bleeding (n = 16, 18.2 %), intracranial hemorrhage (n = 8, 9.1 %), uterine or vaginal bleeding (n = 6, 6.8 %), and pneumorrhagia (n = 6, 6.8 %). Additionally, several rare symptoms were noted, namely white matter lesions in the brain and toxic encephalopathy (n = 3, 3.4 %), splenic infarction (n = 1, 1.1 %), bladder rupture (n = 1, 1.1 %), acute kidney injury (n = 1, 1.1 %), and mediastinal bleeding (n = 1, 1.1 %). Notably, symptom co-occurrence was common, with most patients exhibiting multiple concurrent manifestations.

3.6. Main treatments

All 88 patients received therapeutic intervention, with principal treatments comprising: vitamin K₁ administration (intravenous or oral, n = 83, 94.3 %), fresh frozen plasma infusion (n = 54, 61.4 %), blood transfusion (n = 17, 19.3 %), and prothrombin complex concentrate infusion (n = 17, 19.3 %). These modalities were typically used in combination during clinical management.

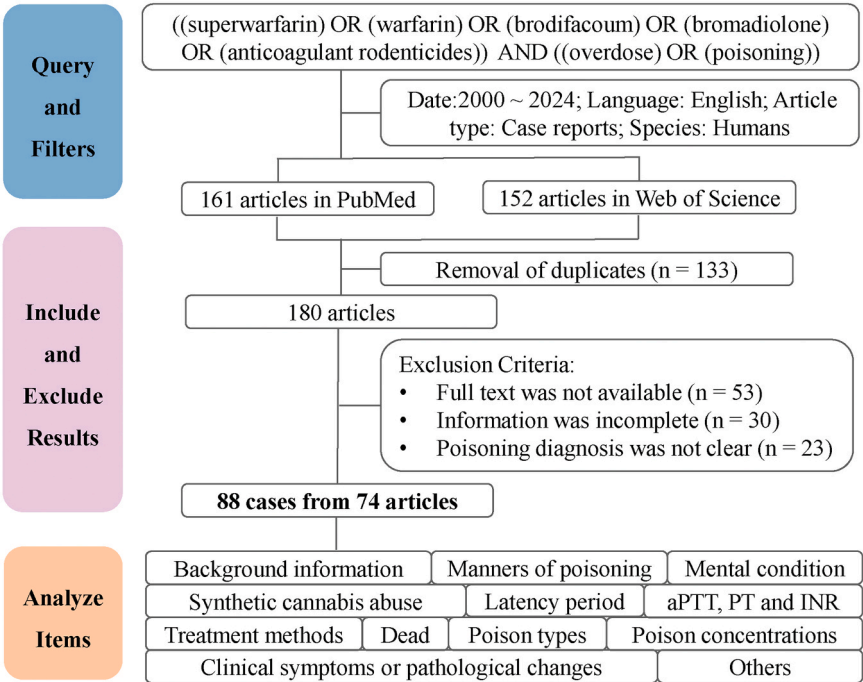


Fig. 1. Process of literature retrieval, screening and analysis.

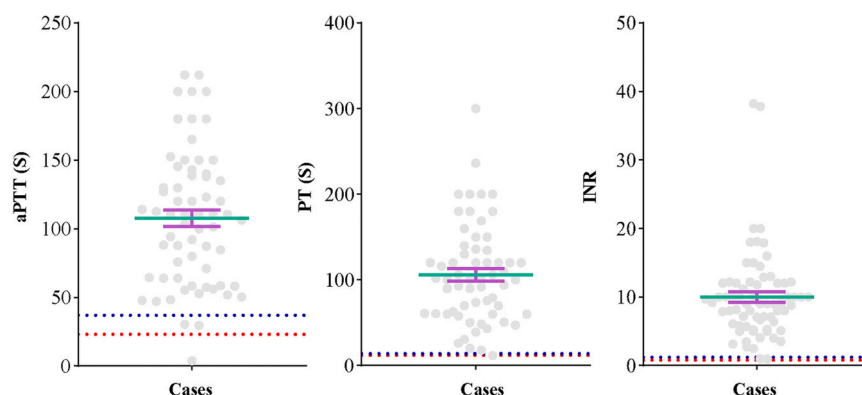


Fig. 2. Results of significant coagulation tests (Note: The green and purple solid lines represent the mean and standard errors, while the blue and red dashed lines represent the upper and lower limits of normal range, respectively).

3.7. Occurrence of death and autopsy

The cohort exhibited a mortality of 6.8 % (6/88 cases). Survival duration among decedents ranged from 0 days (intrauterine death) to 60 days post-exposure, with a mean of 18.17 days and median of 4 days. Forensic autopsy was performed in three fatalities, revealing consistent hemorrhagic pathology: Case 1 [4] demonstrated widespread subcutaneous hemorrhage, subdural and subarachnoid hemorrhage, cerebral parenchymal hemorrhage and edema, pulmonary hemorrhage, edema and emphysema, coronary atherosclerotic plaque hemorrhage, hepatocellular necrosis, glomerular hyalinization, renal tubular epithelial necrosis, and gastric mucosal hemorrhage; Case 2 [47] presented cerebral and pulmonary hemorrhage; Case 3 [64] showed subarachnoid hemorrhage, cerebral parenchymal hemorrhage and edema, multiple subcutaneous hemorrhages, pulmonary edema, and multi-organ congestion. Histological findings confirmed these hemorrhagic and hypoxic alterations.

3.8. Types and concentrations of anticoagulant rodenticides

Toxicological analysis was performed in 35 of the 88 cases (39.8 %), detecting seven distinct rodenticides; two cases involved co-ingestion of multiple rodenticides. Measured concentrations demonstrated substantial variability, ranging from 0.00032 $\mu\text{g/mL}$ to 74.6 $\mu\text{g/mL}$. The distribution of rodenticide types and their corresponding mean concentrations are comprehensively summarized in Table 1.

3.9. Methods of toxicological analysis

Among the 35 cases with quantitatively toxicological analysis, detection methodologies comprised: LC-MS/MS ($n = 22$, 62.9 %), GC-MS ($n = 3$, 8.6 %), HPLC ($n = 1$, 2.9 %), and other unspecified methods ($n = 9$, 25.7 %). Given the predominance of LC-MS/MS in clinical toxicology, we summarized some critical analytical parameters

Table 1
Cases and concentrations of 7 anticoagulant rodenticides in 35 poisoning cases.

Types of rodenticides	Number of cases	Minimum concentration ($\mu\text{g/mL}$)	Maximum concentration ($\mu\text{g/mL}$)	Median ($\mu\text{g/mL}$)
Brodifacoum	17	0.05	10.00	0.15
Bromadiolone	12	0.00032	1.66500	0.12150
Warfarin	4	0.092	74.600	1.919
Chlorophacinone	2	0.042	25.900	NA
Difethialone	1	0.035		
Flocoumafen	1	0.061		
Difenacoum	1	0.037		

Note: "NA": Not applicable

from these reports. Two sample pre-treatment methods were mainly applied: (a) Protein precipitation using acetonitrile with warfarin-d5 as internal standard [4,78,79] and (b) Liquid-liquid extraction using ethyl acetate [5,19,51]. Chromatographic separation on LC-MS/MS were typically performed on a C18 column using mobile phases of: (A) 10 mmol/L ammonium acetate in 5 % methanol and (B) 100 % methanol with gradient elution procedures [53,80,81]. Key mass spectrometric parameters for detection of the seven types of rodenticides were summarized in Table 2.

4. Discussion

Our analysis elucidated key clinical and forensic patterns in anticoagulant rodenticide poisoning. While the predominance of accidental exposure aligns with rodenticides' perceived safety [82], the high suicide rate among psychiatric patients (96.6 %) underscored a critical vulnerability, suggesting targeted prevention strategies were needed. The low homicide incidence (2.3 %) corroborated global trends [83], yet its forensic significance necessitates vigilance in sudden-death investigations.

The extended latency period (median: 4 days) reflected the cumulative mechanism of clotting factor depletion. This delay complicates diagnosis, emphasizing the imperative for routine coagulation monitoring in suspected cases. The characteristically elevated aPTT, PT, and INR values, consistent with profound coagulopathy, provided early diagnostic anchors [41]. Hematuria and multi-organ hemorrhages further signal poisoning, though their non-specificity necessitates toxicological confirmation [84].

Vitamin K₁ remains cornerstone therapy [30], but our data revealed a critical gap: Among the three non-fatal deaths, two did not receive continuous vitamin K₁ treatment, which might be related to the death. This highlights the need for patient adherence protocols, especially given rodenticides' prolonged half-lives [40,74]. The tragic fetal losses in pregnant women (3/3 cases), despite treatment, point to placental transfer of toxins, urging specialized guidelines for this cohort [85,86]. Pathological findings (e.g., hepatic/renal necrosis superimposed on hemorrhage) mechanistically reflected coagulopathic tissue injury [4, 47,64].

Brodifacoum and bromadiolone dominated our toxicological data, reflected their global prevalence and potency. LC-MS/MS emerged as the optimal detection method [51,87,88], yet only 43 % of cases had confirmed toxicology, a limitation stemming from limited lab capabilities. This gap stressed the urgency for accessible toxicology assays, so to facilitate the precise qualitative and quantitative determination of this type of rodenticide, certain methodological parameters were presented.

In summary, this retrospective analysis of 88 anticoagulant rodenticide poisoning cases revealed key forensic characteristics with significant diagnostic and management implications: ① *Manners of poisoning*:

Table 2
Some significant parameters for the detection of the seven types of rodenticides by LC-MS/MS.

Types of rodenticides	Chemical formula	Molecular weight (Da)	Charge state	Precursor/ Product ions (m/z)	Declustering potential (V)	Collision energy (eV)
Brodifacoum	C ₃₁ H ₂₃ BrO ₃	522.08	[M-H] ⁻	521.0/135.0 521.0/187.0	-100	-48 -49
Bromadiolone	C ₃₀ H ₂₃ BrO ₄	526.07	[M-H] ⁻	525.1/250.0 525.1/273.0	-120	-48 -49
Warfarin	C ₁₉ H ₁₆ O ₄	308.10	[M-H] ⁻	307.0/161.0 307.0/250.1	-110	-27 -31
Chlorophacinone	C ₂₃ H ₁₅ ClO ₃	374.07	[M-H] ⁻	373.0/200.9 373.0/145.1	-70	-30 -31
Difethialone	C ₃₁ H ₂₃ BrO ₂ S	538.06	[M-H] ⁻	537.0/151.0 537.0/203.0	-60	-48 -50
Flocoumafen	C ₃₃ H ₂₅ F ₃ O ₄	542.17	[M-H] ⁻	541.1/382.2 541.1/161.1	-85	-37 -45
Difenacoum	C ₃₁ H ₂₄ O ₃	444.17	[M-H] ⁻	443.1/293.3 443.1/134.9	-100	-45 -43

Note: The quantitative ions were **bold-types**. The above parameters are for reference only, and there may be slight differences for different instruments.

Accidental poisoning (59.1 %) was predominant, with suicide (35.2 %) strongly associated with psychiatric comorbidities (96.6 % of suicidal cases). However, while homicide was rare (2.3 %), its possibility necessitates thorough investigation in unexplained hemorrhagic deaths; ②*Diagnostic triad*: The median latency of 4 days and profoundly abnormal coagulation markers (aPTT: 110 s; PT: 100 s; INR: 9) [14] should prompt toxicology testing when coupled with unexplained multi-organ hemorrhage, particularly hematuria, subcutaneous bleeding, gastrointestinal and oral hemorrhage; ③ *Treatment criticalities*: Vitamin K₁ supplementation (94.3 % of cases) remains essential, but treatment duration must extend until rodenticide concentrations were no longer detected or fall below a certain threshold, for example, 0.010 µg/mL for brodifacoum [35,89]. ④ *Pathology and clinical manifestations*: Autopsy findings (multi-organ hemorrhage with hepatic/renal necrosis) directly reflect antemortem coagulopathy, reinforcing the mechanistic link between toxicology and tissue injury; ⑤*Toxicological landscape*: Brodifacoum and bromadiolone dominated confirmed cases, with LC-MS/MS as the optimal detection method.

Nevertheless, insufficient rodenticide concentration data in some studies, likely due to analytical constraints, limited our toxicological correlations. Future work should standardize reporting of compound types, doses, and analytical methods to enhance cross-study comparability.

5. Conclusion

This retrospective analysis conducted on 88 cases of anticoagulant rodenticide poisoning has furnished valuable perspectives regarding the toxicological characteristics inherent in these cases. The outcomes of this study have emphasized the significance of early detection, expeditious treatment, as well as comprehensive toxicological analysis in the diagnosis and management of rodenticide poisoning. In general, this study is capable of serving as a beneficial reference for clinical diagnosis, treatment, and forensic identification in analogous cases. Future research endeavors should center around further exploring the toxicological mechanisms of rodenticides and devising more efficacious treatment strategies.

CRediT authorship contribution statement

Zhonghao Yu: Writing – review & editing, Writing – original draft. **Laijing Sun**: Writing – review & editing, Supervision. **Xiangting Gao**: Resources, Data curation. **Yiwu Zhou**: Writing – review & editing, Supervision, Project administration. **Jiixin Zhang**: Writing – original draft, Visualization. **Yuan Wang**: Project administration, Investigation. **Yuhao Yuan**: Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.forsciint.2025.112660.

Data availability

All data collected for this work were made available in the [Supplementary Material Table S1](#).

References

[1] N. King, M.H. Tran, Long-acting anticoagulant rodenticide (Superwarfarin) poisoning: a review of its historical development, epidemiology, and clinical management, *Transfus. Med. Rev.* 29 (4) (2015) 250–258, <https://doi.org/10.1016/j.tmr.2015.06.002>.
[2] Y.K. Chong, T.W. Mak, Superwarfarin (Long-Acting Anticoagulant Rodenticides) poisoning: from pathophysiology to Laboratory-Guided clinical management, *Clin. Biochem. Rev.* 40 (4) (2019) 175–185, <https://doi.org/10.33176/aacb-19-00029>.
[3] M. Laposata, E.M. Van Cott, M.H. Lev, A.J. Chien, N.L. Harris, L. Ellman, A woman with epistaxis, hematemesis, and altered mental status - multiple vitamin K-dependent coagulation-factor deficiencies due to ingestion of brodifacoum, *N. Engl. J. Med.* 356 (2) (2007) 174–182, <https://doi.org/10.1056/NEJMcpc069032>.
[4] Z. Yu, Y. Yuan, Q. Shi, H. Gao, E. Zhao, J. Ma, Y. Zhou, Systematic forensic identification of a homicide by brodifacoum poisoning: a case report, *J. Forensic Leg. Med.* 108 (2024) 102784, <https://doi.org/10.1016/j.jflm.2024.102784>.
[5] Z. Yu, E. Zhao, Q. Shi, Y. Yuan, J. Ma, L. Zhou, Y. Duan, Y. Zhou, Identification of a murder caused by brodifacoum poisoning based on clinical examinations and LC-MS/MS results, *Int. J. Leg. Med.* 138 (4) (2024) 1323–1328, <https://doi.org/10.1007/s00414-024-03198-z>.
[6] Z. Wang, T. Jiang, M. Mu, C. Shen, Z. Cai, H. Chen, B. Zhang, Small bowel intramural hematoma caused by warfarin: case report and literature review, *Scand. J. Gastroenterol.* 59 (7) (2024) 763–769, <https://doi.org/10.1080/00365521.2024.2337830>.
[7] H. Wang, L. Deng, B. Sun, C. Xing, H. Mao, B. Wu, Bromadiolone May cause severe acute kidney injury through severe disorder of coagulation: a case report, *BMC Nephrol.* 25 (1) (2024) 303, <https://doi.org/10.1186/s12882-024-03746-0>.
[8] F. Hindilerden, E. Aksoy, A.Y. Öztürkmen, G. Türker, E. Gültürk, V.S. Hançer, S. Mercan, Poisoning by butylated hydroxytoluene quinone methide acting as a superwarfarin: first reported case in humans, *Blood Coagul. Fibrinolysis* 35 (5) (2024) 279–281, <https://doi.org/10.1097/mbc.0000000000001307>.
[9] Q.-k He, Y.-h Wu, X.-y Lu, M.-w Liu, Rodenticide poisoning leading to cerebral hemorrhage: a case report, *Medicine* 103 (7) (2024), <https://doi.org/10.1097/md.00000000000036971>.
[10] C.J. Watson, M.D. Simpson, J.D. Whitley, A. Patterson, M.M. Burns, Warfarin overdose in an adolescent not dependent on anticoagulation: reversal strategy and kinetics, *J. Med. Toxicol.* 18 (4) (2022) 334–339, <https://doi.org/10.1007/s13181-022-00907-1>.
[11] A. Setiawan, V. Vohra, E.J. Winograd, Massive warfarin overdose management in an adolescent patient, *Clin. Toxicol. (Phila)* 60 (10) (2022) 1159–1162, <https://doi.org/10.1080/15563650.2022.2116337>.

- [12] A. Lu, F. Yuan, Y. Yao, W. Wen, H. Lu, S. Wu, L. Wang, Reversible leukoencephalopathy caused by 2 rodenticides bromadiolone and fluoroacetate a case report and literature review, *Medicine* 100 (9) (2021), <https://doi.org/10.1097/md.00000000000025053>.
- [13] Q. Li, W. Yu, Y. Qu, J.-Q. Wang, N. Mao, H. Kang, Acute toxic encephalopathy following bromadiolone intoxication: a case report, *Bmc Neurol.* 21 (1) (2021), <https://doi.org/10.1186/s12883-020-02034-2>.
- [14] L. Jia, R. Jia, H.L. Zhang, Convulsive status epilepticus as the initial presentation of superwarfarin poisoning: a case report, *Ann. Palliat. Med.* 10 (6) (2021) 7008–7012, <https://doi.org/10.21037/apm-20-1506>.
- [15] Y. Gou, L. Peng, W. Li, Y. Duan, A rare antinuclear antibody fluorescence pattern induced by rodenticides: a case report, *Clin. Lab* 67 (5) (2021), <https://doi.org/10.7754/Clin.Lab.2020.200933>.
- [16] Y. Tao, T. Liu, J. Han, X. Jian, B. Kan, Clinical characteristics and treatment of mixed-pesticide poisoning in a patient: reflections on a particular case, *J. Int. Med. Res.* 48 (12) (2020) 300060520977392, <https://doi.org/10.1177/0300060520977392>.
- [17] S. Kumar, G. Bhagia, Brodifacoum-Laced synthetic marijuana toxicity: a fight against time, *Am. J. Case Rep.* 21 (2020) e927111, <https://doi.org/10.12659/ajcr.927111>.
- [18] M. Kircher, J. Perez, Brodifacoum poisoning linked to synthetic marijuana use in wisconsin, *Wmj* 119 (2) (2020) 129–131.
- [19] F. Foti, S. Sica, E. Alma, R. De Cristofaro, N. Mores, G. Vetrugno, Sudden nasal bleeding and brodifacoum: a case of accidental exposure or attempted homicide? *Leg. Med. (Tokyo)* 47 (2020) 101772 <https://doi.org/10.1016/j.legalmed.2020.101772>.
- [20] D.Y. Acik, Acquired combined factor deficiency: case report, *Clin. Lab* 66 (9) (2020), <https://doi.org/10.7754/Clin.Lab.2020.191249>.
- [21] W. Zuo, X. Zhang, J.-B. Chang, W.-B. Ma, J.-J. Wei, Bromadiolone poisoning leading to subarachnoid haemorrhage: a case report and review of the literature, *J. Clin. Pharm. Ther.* 44 (6) (2019) 958–962, <https://doi.org/10.1111/jcpt.13005>.
- [22] G. Yilmaz, Splenic infarction after warfarin overdose treatment: is it a coincidence or complication? *Ann. Vasc. Surg.* 59 (2019) 314.e315–314.e317, <https://doi.org/10.1016/j.avsg.2019.02.012>.
- [23] J. Weirthein, D. Scolnik, N.Y. Milshtein, T. Capua, M. Glatstein, Accidental rivaroxaban intoxication in a boy: some lessons in managing new oral anticoagulants in children, *Pedia Emerg. Care* 35 (3) (2019) e44–e46, <https://doi.org/10.1097/pec.0000000000001392>.
- [24] T. Sahin, U. Oner, O. Baser, I. Kurtuncu, Spontaneous bladder rupture secondary to warfarin overdose: a case report, *BMC Emerg. Med.* 19 (1) (2019) 80, <https://doi.org/10.1186/s12873-019-0294-6>.
- [25] C.H. Ross, P. Singh, E.L. Simon, Hemorrhagic soft tissue upper airway obstruction from brodifacoum-contaminated synthetic cannabinoid, *J. Emerg. Med.* 57 (1) (2019) 47–50, <https://doi.org/10.1016/j.jemermed.2019.03.007>.
- [26] S.B. Riley, M. Sochat, K. Moser, K.L. Lynch, R. Tochtrop, T.S. Isbell, A. Scalzo, Case of brodifacoum-contaminated synthetic cannabinoid, *Clin. Toxicol. (Phila)* 57 (2) (2019) 143–144, <https://doi.org/10.1080/15563650.2018.1502444>.
- [27] A. Rasin, J.M. Devgun, D.G. Nosal, T.J. Meehan, R.B. van Breemen, T. M. Thompson, Coagulopathy after synthetic cannabinoid use: a case report, *Ann. Intern. Med.* 171 (8) (2019) 595–596, <https://doi.org/10.7326/119-0321>.
- [28] W. Micula-Gondek, V. Ruales, C. Zorn, A. Angelino, Synthetic cannabinoid-associated coagulopathy: case report, *Psychosomatics* 60 (2) (2019) 211–215, <https://doi.org/10.1016/j.psych.2018.05.010>.
- [29] R.L. Love, N. Sharma, K. Chong, Unusual presentation of haemoptysis following accidental inhalation of the 'superwarfarin' rodenticide brodifacoum, *BMJ Case Rep.* 12 (11) (2019), <https://doi.org/10.1136/bcr-2019-230044>.
- [30] S.C. Simmons, L.J. Taylor, M.B. Marques, L.A. Williams, 3rd, rat poisoning: a challenging diagnosis with clinical and psychological implications, *Lab Med.* 49 (3) (2018) 272–275, <https://doi.org/10.1093/labmed/lmx093>.
- [31] B. Panigrahi, B.C. Jones, S.P. Rowe, Brodifacoum-contaminated synthetic marijuana: clinical and radiologic manifestations of a public health outbreak causing life-threatening coagulopathy, *Emerg. Radiol.* 25 (6) (2018) 715–718, <https://doi.org/10.1007/s10140-018-1628-5>.
- [32] D. Reimer, M. Smith, S. Ali, Deliberate self-poisoning with long-acting anticoagulant rodenticides, *BMJ Case Rep.* 2017 (2017), <https://doi.org/10.1136/bcr-2017-222170>.
- [33] M. Ma, M. Zhang, X. Tang, Z. Li, Massive neonatal intracranial hemorrhage caused by bromadiolone: a case report, *Medicine (Baltimore)* 96 (45) (2017) e8506, <https://doi.org/10.1097/md.00000000000008506>.
- [34] I. Berling, A. Mostafa, J.E. Grice, M.S. Roberts, G.K. Isbister, Warfarin poisoning with delayed rebound toxicity, *J. Emerg. Med.* 52 (2) (2017) 194–196, <https://doi.org/10.1016/j.jemermed.2016.05.068>.
- [35] Y. Wang, V. Kotik, G. Fahim, S. Alagusundaramoorthy, S.A. Eltwansy, S. Mathis, J. Saleh, Treatment of brodifacoum overdose with prothrombin complex concentrate, *Am. J. Health Syst. Pharm.* 73 (1) (2016) e14–e17, <https://doi.org/10.2146/ajhp150233>.
- [36] Z.N. Ozdemir, U. Sahin, M. Merter, M. Gunduz, B. Atesagaoglu, M. Beksac, A case of superwarfarin poisoning due to repetitive occupational dermal rodenticide exposure in a worker, *Turk. J. Hematol.* 33 (3) (2016) 251–253, <https://doi.org/10.4274/tjh.2015.0433>.
- [37] E. Heffler, R. Campisi, S. Ferri, N. Crimi, A bloody mess: an unusual case of diffuse alveolar hemorrhage because of warfarin overdose, *Am. J. Ther.* 23 (5) (2016) e1280–e1283, <https://doi.org/10.1097/mjt.0000000000000397>.
- [38] O. Haesloop, A. Tillick, G. Nichol, J. Strote, Superwarfarin ingestion treated successfully with prothrombin complex concentrate, *Am. J. Emerg. Med.* 34 (1) (2016), <https://doi.org/10.1016/j.ajem.2015.05.033>, 116.e111–112.
- [39] G.S. Booth, P.Z. Mody, Brodifacoum inhalation and its clinical manifestations in a 21-year-old caucasian man, *Lab Med.* 47 (1) (2016) 63–66, <https://doi.org/10.1093/labmed/lmv008>.
- [40] S. Schulman, B. Furie, How I treat poisoning with vitamin K antagonists, *Blood* 125 (3) (2015) 438–442, <https://doi.org/10.1182/blood-2014-08-597781>.
- [41] E.L. Underwood, J. Sutton, I.K. Ellis, B. Qualls, J. Zamber, B.N. Walker, Prolonged coagulopathy after brodifacoum exposure, *Am. J. Health Syst. Pharm.* 71 (8) (2014) 639–642, <https://doi.org/10.2146/ajhp130537>.
- [42] S.S. Matthews, A.L. Ringeisen, B. Wedro, Intentional overdose of warfarin in an adult: anticoagulant reversal in the ED, *Am. J. Emerg. Med.* 32 (9) (2014), <https://doi.org/10.1016/j.ajem.2014.02.025>, 1150.e1151–1152.
- [43] A. Kogan, R. Aitchison, D.S. Lees, C.L. Landoli, K. Dayrit, M. Kharasch, E. Wang, 61-year-old man with sudden onset of gross hematuria: a case report, *Am. J. Ther.* 21 (1) (2014) E7–E8, <https://doi.org/10.1097/MJT.0b013e31825fa0e6>.
- [44] S. Keskin, I.E. Gökmen, O. Koç, O. Özbek, Unusual complication of coumadin toxicity mimicking Crohn's disease, *BMJ Case Rep.* 2014 (2014), <https://doi.org/10.1136/bcr-2013-202755>.
- [45] D.J. Card, S. Francis, K. Deuchande, D.J. Harrington, Superwarfarin poisoning and its management, *BMJ Case Rep.* 2014 (2014), <https://doi.org/10.1136/bcr-2014-206360>.
- [46] A. Zammit, D.G. Marguerat, C. Caruana, Anticoagulation-induced spontaneous intramural small bowel haematomas, *BMJ Case Rep.* 2013 (2013), <https://doi.org/10.1136/bcr-2013-008831>.
- [47] J. Yan, Y. Shi, C. Sun, H. Yang, Vitamin K treatment of brodifacoum poisoning in a pregnant woman, *Int. J. Gynaecol. Obstet.* 122 (2) (2013) 162–163, <https://doi.org/10.1016/j.ijgo.2013.03.008>.
- [48] S. Rutović, M. Dikanović, I. Mirković, G. Lojen, M. Marčić, Intracerebellar hemorrhage caused by superwarfarin poisoning, *Neurol. Sci.* 34 (11) (2013) 2071–2072, <https://doi.org/10.1007/s10072-013-1445-2>.
- [49] K.M. Mehlhaff, C.C. Baxter, K. Rudinsky, D.S. McKenna, Lethal neonatal coagulopathy after maternal ingestion of a superwarfarin, *Obstet. Gynecol.* 122 (2 Pt 2) (2013) 500–502, <https://doi.org/10.1097/AOG.0b013e31829267c4>.
- [50] D. Franco, G. Everett, M. Manoucheri, I smell a rat: a case report and literature review of paradoxical thrombosis and hemorrhage in a patient with brodifacoum toxicity, *Blood Coagul. Fibrinolysis* 24 (2) (2013) 202–204, <https://doi.org/10.1097/MBC.0b013e328358e959>.
- [51] H. Yan, P. Xiang, L. Zhu, M. Shen, Determination of bromadiolone and brodifacoum in human blood using LC-ESI/MS/MS and its application in four superwarfarin poisoning cases, *Forensic Sci. Int.* 222 (1–3) (2012) 313–317, <https://doi.org/10.1016/j.forsciint.2012.07.008>.
- [52] M.B. Centorino, D.L. Sanchez, G. Catalano, M.C. Catalano, The use of bromadiolone to potentiate self-mutilatory bleeding: a case report, *Psychosomatics* 53 (5) (2012) 489–493, <https://doi.org/10.1016/j.psych.2012.01.009>.
- [53] N. Gunja, A. Coggins, S. Bidny, Management of intentional superwarfarin poisoning with long-term vitamin K and brodifacoum levels, *Clin. Toxicol. (Phila)* 49 (5) (2011) 385–390, <https://doi.org/10.3109/15563650.2011.587126>.
- [54] S. Boettcher, A. Wacker, K. Moerike, H.G. Kopp, K. Jaschonek, T. Grobosch, L. Kanz, H.R. Salih, Acquired coagulopathy caused by intoxication with the superwarfarin-type anticoagulant rodenticide flocoumafen, *Eur. J. Haematol.* 86 (2) (2011) 173–175, <https://doi.org/10.1111/j.1600-0609.2010.01550.x>.
- [55] S.-L. Zhao, P. Li, M. Ji, Y. Zong, S.-T. Zhang, Upper gastrointestinal hemorrhage caused by superwarfarin poisoning, *World J. Gastroenterol.* 16 (13) (2010) 1680–1682, <https://doi.org/10.3748/wjg.v16.i13.1680>.
- [56] H. Nie, W. Wei, Y. Cao, X.D. Du, Superwarfarin poisoning as a cause of intestinal obstruction, *Clin. Toxicol. (Phila)* 48 (3) (2010) 238–239, <https://doi.org/10.3109/15563650903555302>.
- [57] S.Y. Kim, S.Y. Cho, H.J. Lee, J.-T. Suh, S.H. Oh, W.-I. Lee, T.S. Park, H.-J. Yoon, Superwarfarin intoxication of unknown etiology accompanying hemoperitoneum in a patient on fluconazole therapy, *Ann. Clin. Lab. Sci.* 40 (3) (2010) 300–303.
- [58] E.V. De Paula, S.A. Montalva, P.R. Madureira, R. Jose Vieira, J.M. Annichino-Bizzacchi, M.C. Ozelo, Simultaneous bleeding and thrombosis in superwarfarin poisoning, *Thromb. Res.* 123 (4) (2009) 637–639, <https://doi.org/10.1016/j.thromres.2008.02.004>.
- [59] M. Zolcinski, A. Padjas, J. Musial, Intoxication with three different superwarfarin compounds in an adult woman, *Thromb. Haemost.* 100 (1) (2008) 156–157, <https://doi.org/10.1160/th08-03-0132>.
- [60] M. Serinken, O. Karcioğlu, B. Yagci, Intramuscular hemorrhage resulting in shock associated with warfarin toxicity, *Clin. Toxicol. (Phila)* 46 (8) (2008) 765–767, <https://doi.org/10.1080/15563650701639725>.
- [61] V.M.H. Lo, C.K. Ching, A.Y.W. Chan, T.W.L. Mak, Bromadiolone toxicokinetics: diagnosis and treatment implications, *Clin. Toxicol.* 46 (8) (2008) 703–710, <https://doi.org/10.1080/15563650701504366>.
- [62] S.-W. Hou, C.-C. Chen, K.-C. Chen, S.-Y. Ko, C.-S. Wong, C.-F. Chong, Sonographic diagnosis of spontaneous intramural small bowel hematoma in a case of warfarin overdose, *J. Clin. Ultrasound* 36 (6) (2008) 374–376, <https://doi.org/10.1002/jcu.20442>.
- [63] J.E. Spahr, J.S. Maul, G.M. Rodgers, Superwarfarin poisoning: a report of two cases and review of the literature, *Am. J. Hematol.* 82 (7) (2007) 656–660, <https://doi.org/10.1002/ajh.20784>.
- [64] F. Papin, F. Clarot, C. Vicomte, J.M. Gaulier, C. Daubin, F. Chapon, E. Vaz, B. Proust, Lethal paradoxical cerebral vein thrombosis due to suspicious anticoagulant rodenticide intoxication with chlorophacinone, *Forensic Sci. Int.* 166 (2–3) (2007) 85–90, <https://doi.org/10.1016/j.forsciint.2006.04.003>.
- [65] V. Olmos, C. Magdalena Lopez, Brodifacoum poisoning with toxicokinetic data, *Clin. Toxicol.* 45 (5) (2007) 487–489, <https://doi.org/10.1080/15563650701354093>.

- [66] M. Laposata, E.M. Van Cott, M.H. Lev, Case records of the massachusetts general hospital. Case 1-2007. A 40-year-old woman with epistaxis, hematemesis, and altered mental status, *N. Engl. J. Med.* 356 (2) (2007) 174–182, <https://doi.org/10.1056/NEJMcp069032>.
- [67] P. Kotsaftis, F. Girtovitis, A. Boutou, G. Ntaios, P.E. Makris, Haemarthrosis after superwarfarin poisoning, *Eur. J. Haematol.* 79 (3) (2007) 255–257, <https://doi.org/10.1111/j.1600-0609.2007.00904.x>.
- [68] S. Binks, P. Davies, Case of the month: "oh! Drat!—A case of transcutaneous superwarfarin poisoning and its recurrent presentation", *Emerg. Med. J.* 24 (4) (2007) 307–308, <https://doi.org/10.1136/emj.2006.044925>.
- [69] M.A. Miller, P.D. Levy, D. Hile, Rapid identification of surreptitious brodifacoum poisoning by analysis of vitamin K-dependent factor activity, *Am. J. Emerg. Med.* 24 (3) (2006) 383, <https://doi.org/10.1016/j.ajem.2005.11.009>.
- [70] T. Grobosch, B. Angelow, L. Schönborg, D. Lampe, Acute brodifacoum intoxication, *J. Anal. Toxicol.* 30 (4) (2006) 281–286, <https://doi.org/10.1093/jat/30.4.281>.
- [71] S. Zupancić-Salek, J. Kovacević-Metelko, I. Radman, Successful reversal of anticoagulant effect of superwarfarin poisoning with recombinant activated factor VII, *Blood Coagul. Fibrinolysis* 16 (4) (2005) 239–244, <https://doi.org/10.1097/01.mbc.0000169215.70184.56>.
- [72] S. Sarin, H. Mukhtar, M.A. Mirza, Prolonged coagulopathy related to superwarfarin overdose, *Ann. Intern. Med.* 142 (2) (2005) 156, <https://doi.org/10.7326/0003-4819-142-2-200501180-00024>.
- [73] J. Pavlu, D.J. Harrington, K. Voong, G.F. Savidge, R. Jan-Mohamed, R. Kaczmariski, Superwarfarin poisoning, *Lancet* 365 (9459) (2005) 628, [https://doi.org/10.1016/s0140-6736\(05\)17916-1](https://doi.org/10.1016/s0140-6736(05)17916-1).
- [74] B.T. Tsutaoka, M. Miller, S.M. Fung, M.M. Patel, K.R. Olson, Superwarfarin and glass ingestion with prolonged coagulopathy requiring high-dose vitamin K1 therapy, *Pharmacotherapy* 23 (9) (2003) 1186–1189, <https://doi.org/10.1592/phco.23.10.1186.32755>.
- [75] A.V. Ramanan, P. Gissen, B. Bose-Haider, Intentional overdose of warfarin in an adolescent: need for follow up, *Emerg. Med. J.* 19 (1) (2002) 90, <https://doi.org/10.1136/emj.19.1.90>.
- [76] G.R. Bruno, M.A. Howland, A. McMeeking, R.S. Hoffman, Long-acting anticoagulant overdose: brodifacoum kinetics and optimal vitamin k dosing, *Ann. Emerg. Med.* 36 (3) (2000) 262–267, <https://doi.org/10.1067/mem.2000.108317>.
- [77] D. Bates, M. Mintz, Phytonadione therapy in a multiple-drug overdose involving warfarin, *Pharmacotherapy* 20 (10) (2000) 1208–1215, <https://doi.org/10.1592/phco.20.10.1208.34584>.
- [78] A. Arens, J. Teske, M. Klitschar, R. Mischke, M. Dziadosz, Antemortem and postmortem rodenticide analysis in forensic toxicology as a part of an LC-MS/MS-based multi-target screening strategy, *Drug Test. Anal.* 14 (6) (2022) 1149–1154, <https://doi.org/10.1002/dta.3222>.
- [79] H. Guo, J. Wang, Y. Wu, W. Liu, J. Bu, Q. Zhao, Sensitive and simultaneous determination of nine anticoagulant rodenticides in human blood by UPLC-MS-MS with phospholipid removal pretreatment, *J. Anal. Toxicol.* 42 (7) (2018) 459–466, <https://doi.org/10.1093/jat/bky024>.
- [80] L. Zhu, H. Yan, B. Shen, Y. Shi, M. Shen, P. Xiang, Determination of brodifacoum and brodifacoum in human hair by liquid chromatography/tandem mass spectrometry and its application to poisoning cases, *Rapid Commun. Mass Spectrom.* 27 (4) (2013) 513–520, <https://doi.org/10.1002/rcm.6477>.
- [81] Z. Yan, H. Li, H. Li, G. Lai, J. Chu, H. Guo, Q. Zhao, Simultaneous determination of nine anticoagulant rodenticides by ultra-performance liquid chromatography-tandem mass spectrometry with ultrasound-assisted low-density solvent dispersive liquid-liquid microextraction, *J. Chromatogr. B Anal. Technol. Biomed. Life Sci.* 1092 (2018) 453–458, <https://doi.org/10.1016/j.jchromb.2018.06.053>.
- [82] D.L. Feinstein, B.S. Akpa, M.A. Ayee, A.I. Boullerne, D. Braun, S.V. Brodsky, D. Didalevitz, Z. Hauck, S. Kalinin, K. Kowal, I. Kuzmenko, K. Lis, N. Marangoni, M. W. Martynowycz, I. Rubinstein, R. van Breemen, K. Ware, G. Weinberg, The emerging threat of superwarfarins: history, detection, mechanisms, and countermeasures, *Ann. N. Y. Acad. Sci.* 1374 (1) (2016) 111–122, <https://doi.org/10.1111/nyas.13085>.
- [83] M.T. Lohr, R.A. Davis, Anticoagulant rodenticide use, non-target impacts and regulation: a case study from Australia, *Sci. Total Environ.* 634 (2018) 1372–1384, <https://doi.org/10.1016/j.scitotenv.2018.04.069>.
- [84] Y.F. Wu, C.S. Chang, C.Y. Chung, H.Y. Lin, C.C. Wang, M.C. Shen, Superwarfarin intoxication: hematuria is a major clinical manifestation, *Int. J. Hematol.* 90 (2) (2009) 170–173, <https://doi.org/10.1007/s12185-009-0374-6>.
- [85] D. Shan, Y. Ji, Y. Hu, T. Li, Treasure to the mother and threat to the fetus: case report of warfarin-associated fetal intracranial hemorrhage and review of literature, *J. Int. Med. Res.* 51 (8) (2023) 3000605231192773, <https://doi.org/10.1177/03000605231192773>.
- [86] M. Yurdakök, Fetal and neonatal effects of anticoagulants used in pregnancy: a review, *Turk. J. Pediatr.* 54 (3) (2012) 207–215.
- [87] D.G. Nosal, D.L. Feinstein, L. Chen, R.B. van Breemen, Separation and quantification of superwarfarin rodenticide Diastereomers-Bromadiolone, difenacoum, flocoumafen, brodifacoum, and Difethialone-in human plasma, *J. AOAC Int.* 103 (3) (2020) 770–778, <https://doi.org/10.1093/jaoacint/qsaa007>.
- [88] A.M. Hernández, J. Bernal, J.L. Bernal, M.T. Martín, C. Caminero, M.J. Nozal, Simultaneous determination of nine anticoagulant rodenticides in soil and water by LC-ESI-MS, *J. Sep. Sci.* 36 (16) (2013) 2593–2601, <https://doi.org/10.1002/jssc.201300310>.
- [89] L. Yip, N.V. Stanton, R.A. Middleberg, Vitamin K(1) treatment duration in patients with brodifacoum poisoning, *N. Engl. J. Med.* 382 (18) (2020) 1764–1765, <https://doi.org/10.1056/NEJMc1916199>.